Fitness Testing 1: Study Notes

Course: Personal Training Certification

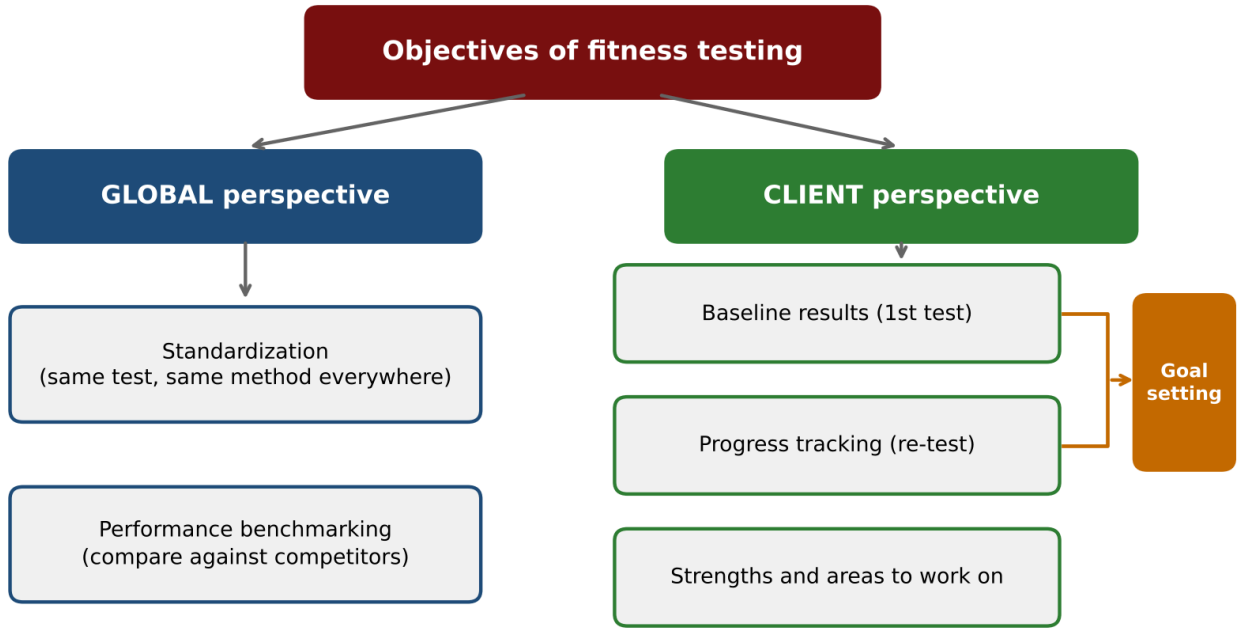
Topic: Fitness testing, part 1 of 2

About these notes: A short revision version of the Fitness Testing 1 lecture: key tables, key terms and 20 revision questions with an answer key. For the full explanation see the Detailed Notes. Sources: the lecture, the course slides ("Fitness Testing 1 & 2"), board photos from the lecture and E-book Chapter VII (Health Screening and Fitness Testing).

Big idea: Fitness is tested with standardised, worldwide tests. Before any test, screen the client's health. Then test in an order that protects the accuracy of every result: blood pressure, body composition, cardiorespiratory endurance, muscular strength, muscular endurance, flexibility. Body composition has eight tests, and six of them (waist circumference, waist to hip ratio, BMI, skinfold, BIA and Bod Pod) are covered today.

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1. Why test? Objectives of fitness testing



Strengths and areas to work on also drive the customised workout plan

Perspective	Objective	Meaning
Global	**Standardisation**	Same test, procedure, method and norms everywhere (for example bench press is the worldwide muscular strength test), so results can be compared
Global	**Performance benchmarking**	Shows how far a country's athletes lag or lead, so the standard can be chased or maintained
Client	**Baseline results**	First test: where the client stands today
Client	**Progress tracking**	Re-test (for example after 3 months): numbers show progress, build confidence and motivation
Client	**Goal setting**	From baseline and re-test: short-term, mid-term and long-term milestones, tweaked after the re-test
Client	**Strengths and areas to work on**	Customised plan (say "areas to work on", not "weaknesses"); strength can come from any of the five components

- A **test norm** is a sheet that shows where a score stands for a component.
- All five components must be tested before calling someone fit.
- A **contraindication** is any movement that causes discomfort or pain: choose an alternative exercise to avoid injury.
- Poor flexibility result: stretch every day (mobility drills and static stretching).

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2. Health screening (pre-step)

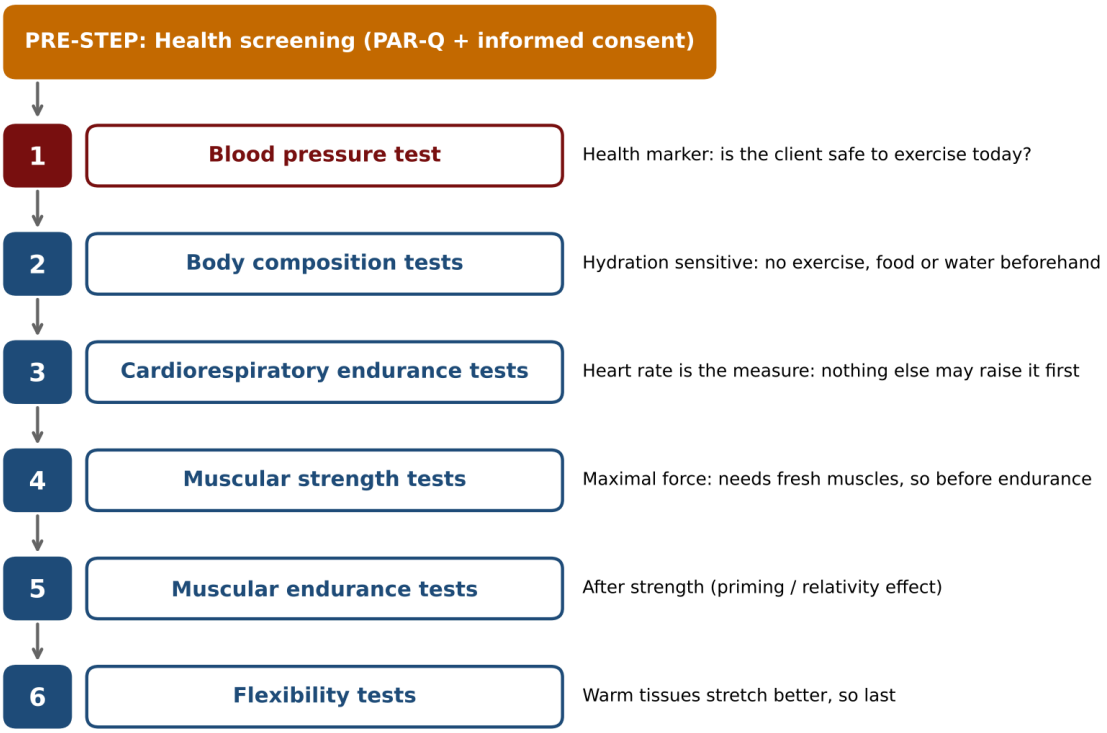
Health screening is a procedure done on people without symptoms to assess the likelihood of a disease. Think of baggage screening at an airport: make sure the client is safe before testing. It uses two forms, filled and signed by the client.

Form	Key points
PAR-Q (Physical Activity Readiness Questionnaire)	About 10 close-ended Yes or No questions on medical background (heart, joints, neurological). **All No** means cleared. Any **Yes** means stop, ask, and get a **NOC** (no objection certificate) from the doctor
Informed consent	A script that informs the client about machines, exercises (strength, cardio, stretching) and risks, for the whole fitness journey, ending with a statement of willing agreement and a signature

- Document everything and get it signed: it protects the trainer.
- If a gym already has its own forms, follow them. If it has none, introduce them.

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3. Order of fitness testing



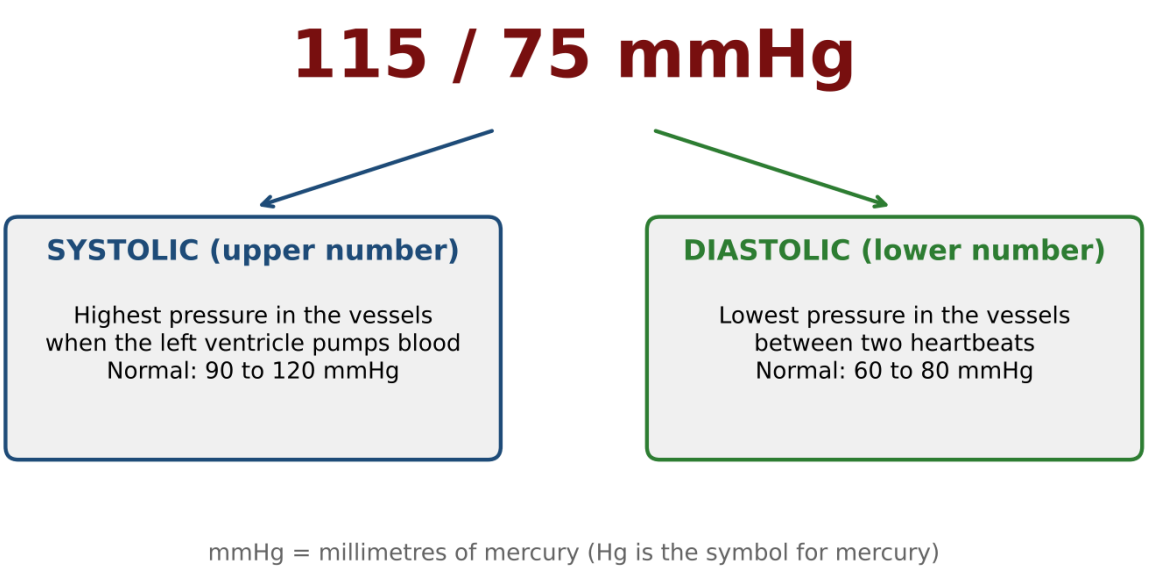
Steps 2 to 6 are the five components of fitness. The order is for accurate results, not a ranking of importance.

Step	Test	Why here
1	Blood pressure	Health marker for the day; high BP could shoot up during testing
2	Body composition	Needs a neutral body (exercise, sweat, food and water change body weight and composition)
3	Cardiorespiratory endurance	Heart rate must rise only from the cardio task
4	Muscular strength	Maximum force in one effort needs fresh muscles
5	Muscular endurance	After strength: priming or relativity effect (the 100 km/h feels slower after 200 km/h)
6	Flexibility	Tissues are warm after steps 3 to 5, so they lengthen better

- The order is **not** a ranking of importance.
- **Muscular strength:** maximum force in one single effort. **Muscular endurance:** submaximal contraction for a long time without fatigue.
- **Mobility** (joints moving freely and actively through full range of motion without discomfort) is tested side by side with flexibility.

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4. Test 1: blood pressure



mmHg = millimetres of mercury (Hg is the symbol for mercury)

- **Systolic** (upper): highest pressure, when the left ventricle pumps. Normal 90 to 120 mmHg.
- **Diastolic** (lower): lowest pressure, between heartbeats. Normal 60 to 80 mmHg.
- **mmHg** is millimetres of mercury.

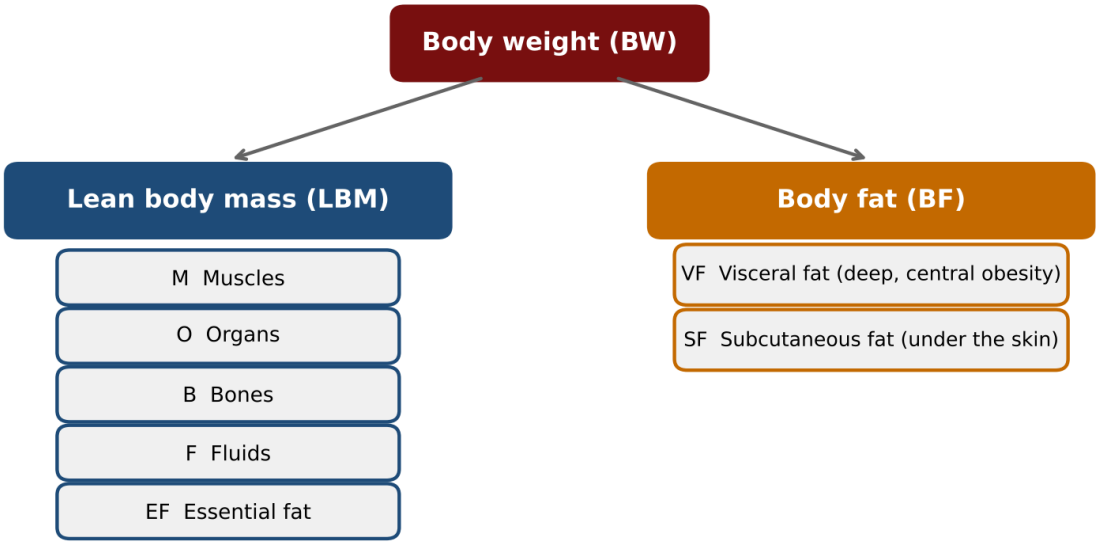
Category	Systolic		Diastolic
Low (hypotension)	Less than 90	Or	Less than 60
Normal	90 to 120	And	60 to 80
Elevated	120 to 129	And	Less than 80
Hypertension stage 1	130 to 139	Or	80 to 89
Hypertension stage 2	140 or higher	Or	90 or higher
Hypertensive crisis	180 or higher	And/or	Higher than 120

Seven steps with a digital monitor: 1 plug the tube into the device; 2 cuff on the upper arm, tube and palm facing up; 3 cuff edge 2 to 3 cm above the elbow; 4 snug (two fingers fit); 5 tube along the medial side of the arm; 6 cuff at heart height (table or armrest); 7 legs uncrossed, feet flat, back supported, no talking.

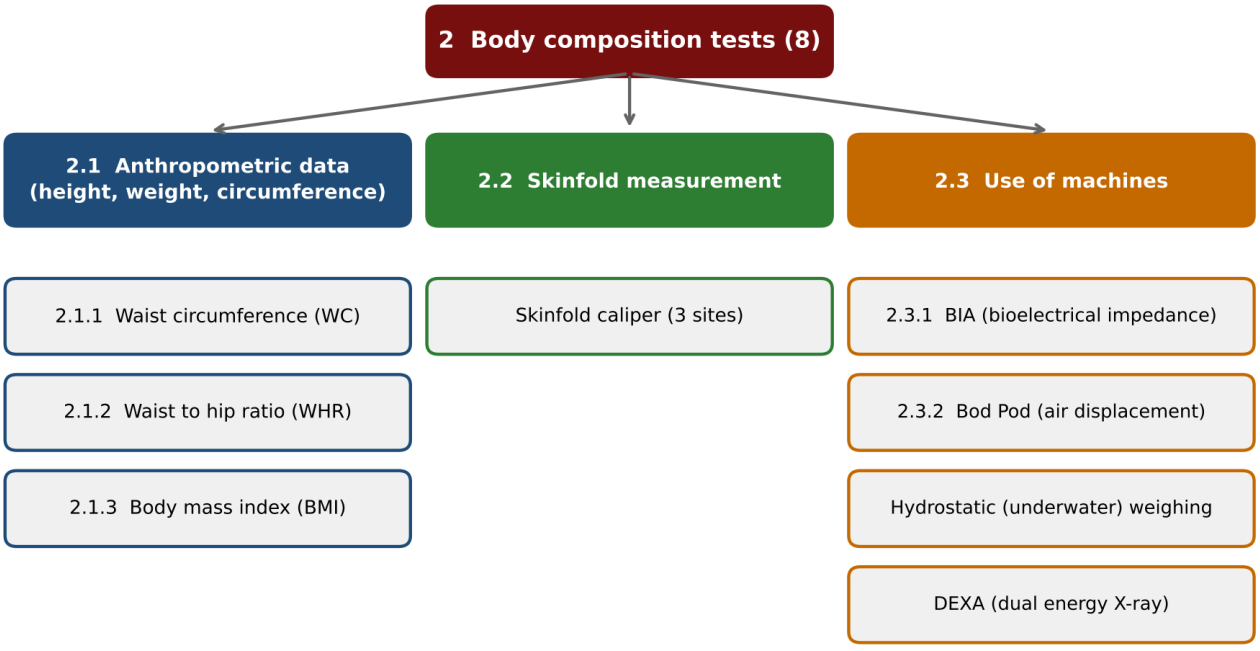
If BP is high: rest, offer water, re-check. If still high, skip the test and try the next day, up to three days. High on all three days: refer to a doctor.

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5. Body composition: the eight tests



Body composition = the ratio of lean body mass to body fat



Lecture 1 covers 2.1, 2.2 and the first two machines; the last two are in Fitness Testing 2

Category	Tests
2.1 Anthropometric data	WC, WHR, BMI
2.2 Skinfold measurement	Skinfold caliper
2.3 Use of machines	BIA, Bod Pod (today); hydrostatic weighing and DEXA (Fitness Testing 2)

- **LBM** = muscles, organs, bones, fluids, essential fat (M, O, B, F, EF). **BF** = visceral fat and subcutaneous fat.

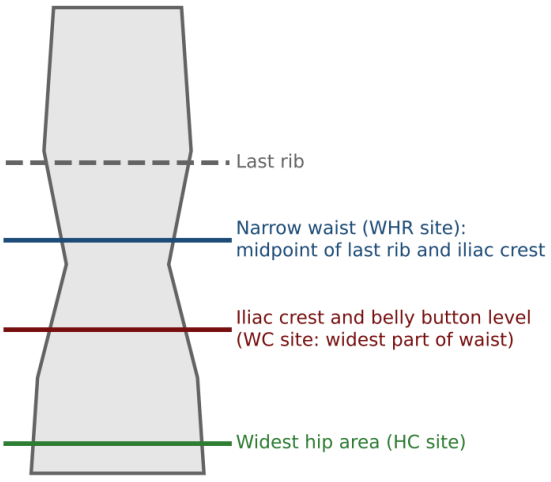
- **Anthropometry** = study of measurements and proportions of the human body (height with a stadiometer, weight with a scale, circumference with a tape).
- Body composition is tested **first** (after BP) because weight and fluid change after activity. BIA is best done first thing in the morning.

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6. Anthropometric tests (2.1)

Test	Objective	Site / formula	Norm (low risk)
WC	Health risk from **visceral fat** (central obesity)	Across the **iliac crest**, through the belly button (widest part of the waist)	Men below **35.5 in (90 cm)**, women below **31.5 in (80 cm)**
WHR	**Overall** health risk	WHR = WC / HC. Waist: narrowest part (midpoint between last rib and iliac crest). Hip: widest part	Men below **0.95**, women below **0.86**
BMI	Category: underweight, normal, overweight, obese	BMI = weight (kg) / height (m)². Unit kg/m²	See table below

Measuring sites (schematic, not to scale)



Test	Male	Female
WC (low risk)	< 35.5 in (90 cm)	< 31.5 in (80 cm)
WHR (low risk)	< 0.95	< 0.86

WHR = waist circumference / hip circumference
(use the same unit for both)

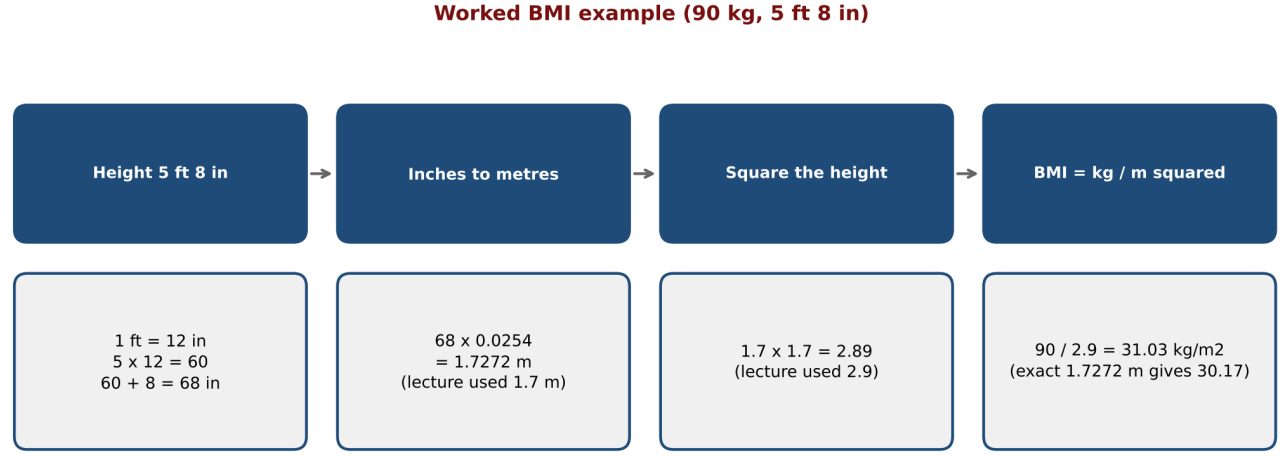
Case: waist 36 in, hip 40 in
WHR = 36 / 40 = 0.90
Male: 0.90 < 0.95 -> low overall risk
Female: 0.90 > 0.86 -> high overall risk

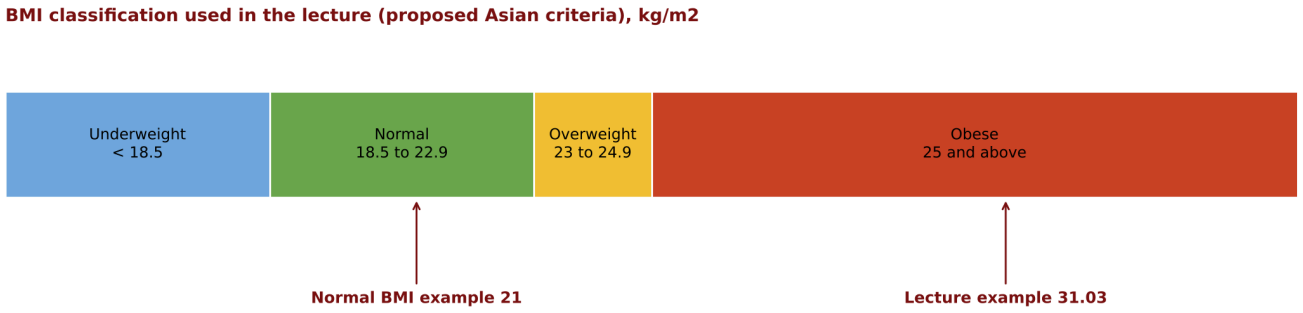
- Ask permission before measuring (or let the client place the tape). Say "widest part of your waist", not "iliac crest".
- WHR case: waist 36 in, hip 40 in gives 0.90. For a man that is low risk (below 0.95). For a woman it is high risk (above 0.86). Use the same unit for waist and hip.
- WC and WHR give risk only, not a body fat percentage.

BMI classification (Asian criteria, slides):

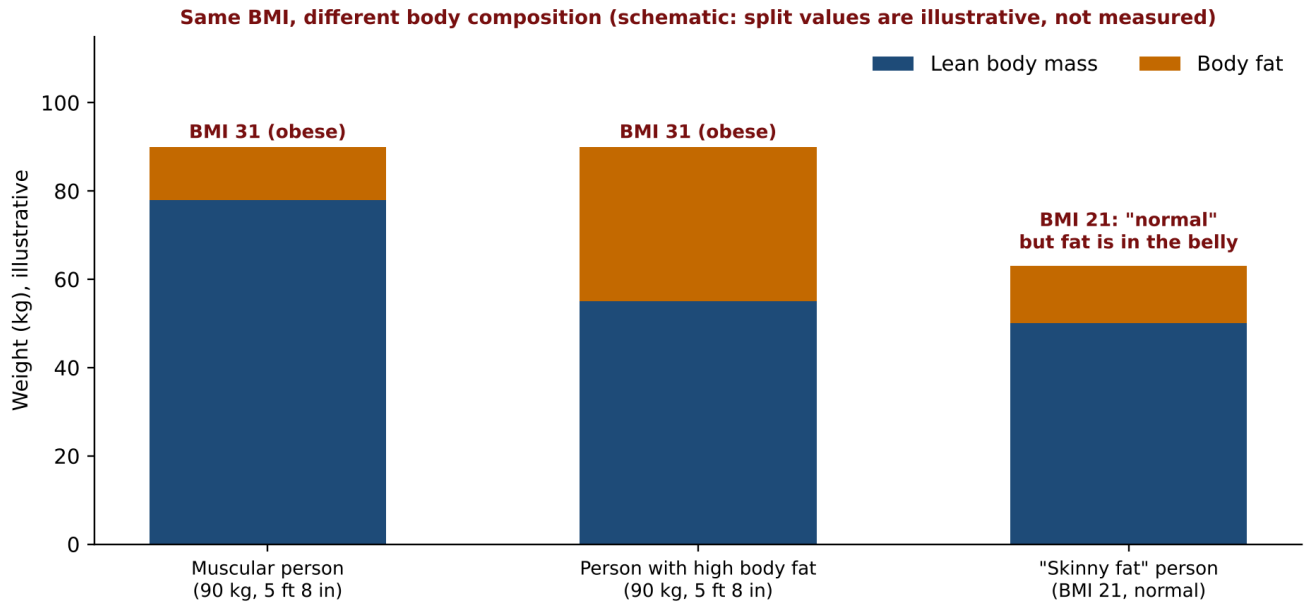
Category	BMI (kg/m²)
Underweight	Below 18.5
Normal	18.5 to 22.9
Overweight	23 to 24.9
Obese	25 and above

BMI calculation: 5 ft 8 in = 5 × 12 + 8 = 68 in. Multiply by 0.0254 to get metres (68 × 0.0254 = 1.7272 m; the board used 1.7 m). Lecture example: 90 / (1.7 × 1.7) = 90 / 2.9 = 31.03 kg/m², obese.





Shortcomings of BMI: it does not separate body fat from lean body mass (a muscular person and a fat person of the same height and weight get the same BMI), and it misses "skinny fat" people (BMI normal, for example 21, but fat in the belly).



- Ideal body fat: men below 15%, women below 20%. None of the three tests gives this percentage.
- Use them only as short-term or mid-term goals, not to judge body composition.

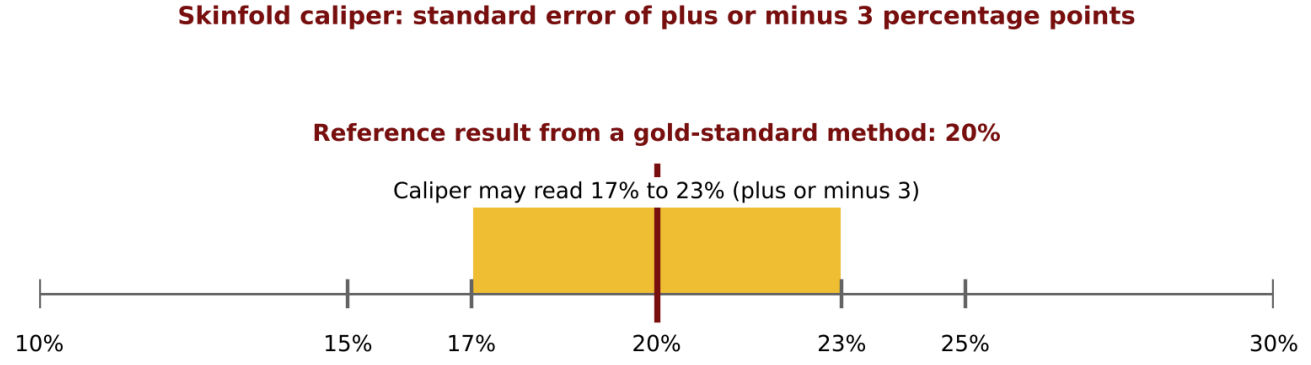
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7. Skinfold caliper test (2.2)

- A compass-like caliper measures the thickness of a pinched **skin fold**, which is **subcutaneous fat**. More subcutaneous fat means a higher body fat percentage.

	Women	Men
Sites	Triceps, suprailiac (just above the iliac crest), mid thigh	Chest (midpoint of armpit top and nipple), abdomen (1 inch from the belly button), mid thigh

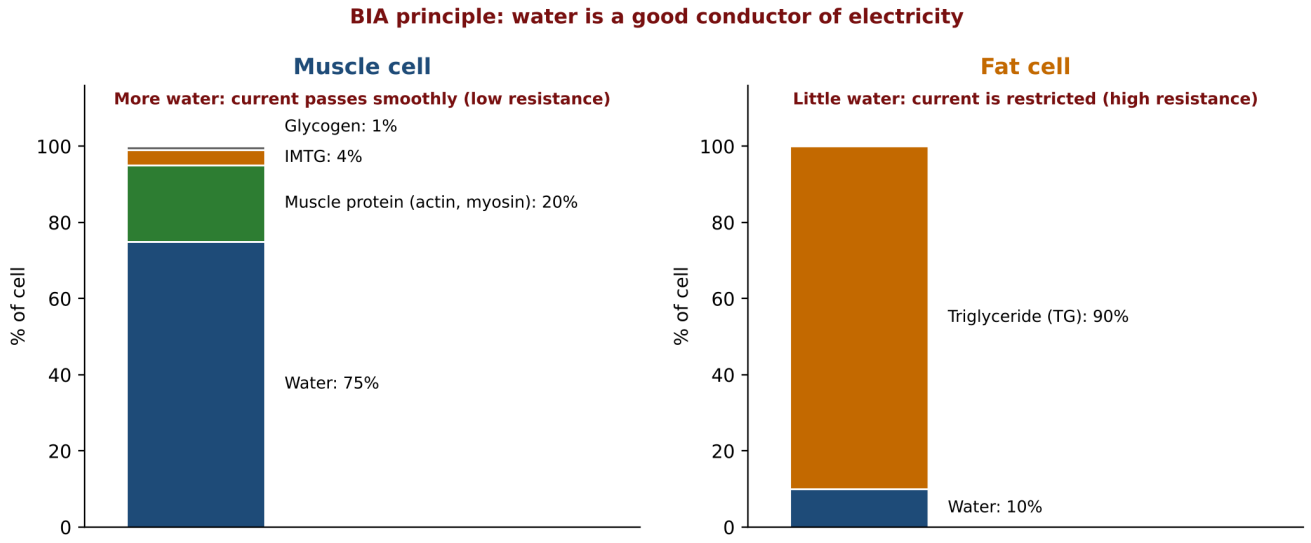
Advantages	Disadvantages
Portable, easy to use, time saving	Only subcutaneous fat (no visceral), standard error of plus or minus 3%, not for the very obese or very lean, pinching discomfort



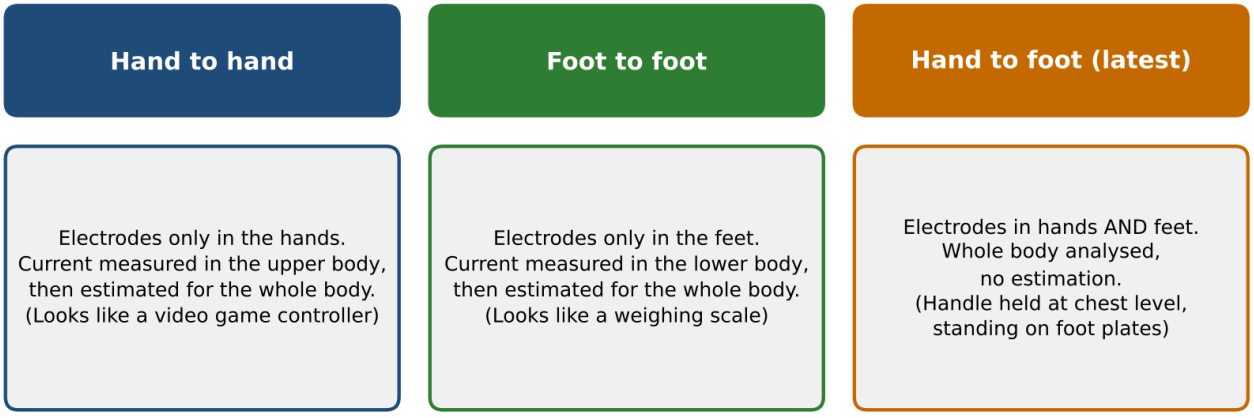
- Price (about Rs 300 to Rs 5,000) does not remove the error. A reference of 20% can read 17% to 23%.

8. BIA test (2.3.1)

- **Bioelectrical Impedance Analysis:** bio (living organism), electric (current), impedance (resistance), analysis (study). Often wrongly called the "BMI machine" (it also reports BMI).
- **Principle:** water is a good conductor of electricity. A mild current enters through electrodes (metal plates); the machine measures resistance.



	Muscle	Fat
Water	75%	10%
Current	Passes smoothly	Restricted



- Types: **hand to hand** (upper body, then estimated), **foot to foot** (lower body, then estimated), **hand to foot** (latest, whole body). Price about Rs 10,000 to Rs 45 lakh; high-end models give segmental analysis and intracellular and extracellular water.
- Results: body fat %, muscle mass, BMI.

Before a BIA test

Rested and empty stomach	No water, no food (food contains water), no coffee
No exercise: 12 hours	Sweat removes fluid (dehydrates)
No alcohol: 48 hours	Effect must be neutralised first
No metal objects	Jewellery, bracelets, watches, phones, belts conduct current
Women: not during periods	Fluid loss changes the reading

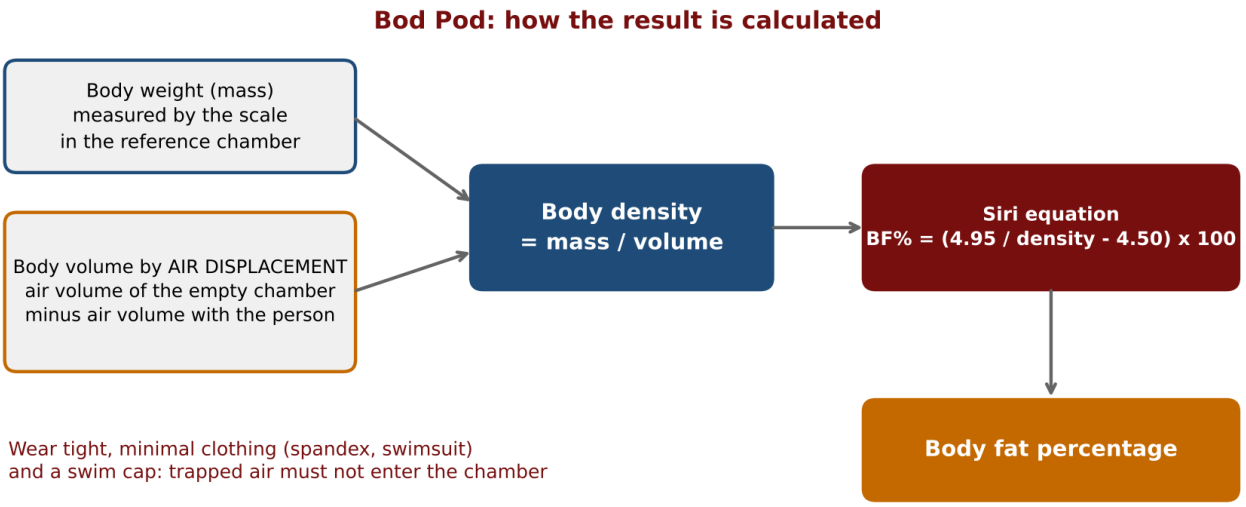
Goal: neutral hydration, with nothing added and nothing lost

Advantages	Disadvantages
Time saving (about a minute), easy to use (even front desk staff), needs only a switch and socket	Some models show standard error (not as accurate as the gold standard), hydration level changes results

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9. Bod Pod test (2.3.2)

- **Bod** = body, **Pod** = sealed chamber (egg-shaped). **Air displacement method**, a **densitometry** method.
- **Body density = body mass / body volume.** Volume = air in the empty chamber minus air with the person. A diaphragm measures air volume; a scale in the reference chamber measures weight.
- **Siri equation:** body fat % = (4.95 / body density minus 4.50) x 100 (no need to memorise).
- Wear spandex or a swimsuit and a tight swim cap so no extra air enters. A breathing tube helps account for lung air.



Advantages	Disadvantages
Accurate (comparable to, but not, the gold standard), time saving, no technical expertise, suits all ages, patients, obese, children, elderly and hydrophobic people	Expensive, limited availability (not available in India, as on the board)

- Example in the video: 97 kg total = 82 kg fat-free mass + 15 kg fat mass (about 15.5% fat, calculated for these notes).
- **Affordability and availability** are also criteria when the best test is chosen in Fitness Testing 2.

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10. Key terms

Term	Meaning
Test norm	Reference sheet showing where a test score stands
Standardisation	Same test, method and norms worldwide
Benchmarking	Comparing performance against others to find the gap
Baseline	Results of the first test
Contraindication	Movement that causes discomfort or pain
Health screening	Check before testing that the client is safe
PAR-Q	Physical Activity Readiness Questionnaire (Yes or No questions)
NOC	Doctor's no objection certificate
Informed consent	Script plus signed agreement to take part
Systolic / diastolic	Highest / lowest pressure in the vessels
Hypotension / hypertension	Low / high blood pressure
Anthropometry	Study of body measurements and proportions
Stadiometer	Device to measure height
Visceral fat	Deep fat around the organs (central obesity)
Subcutaneous fat	Fat under the skin
WC / WHR / BMI	Waist circumference / waist to hip ratio / body mass index
Skinfold caliper	Compass-like device that measures a pinched skin fold
Standard error	The range a result can differ from the true value (for example plus or minus 3%)
Gold standard	The reference test other tests are compared with
BIA	Bioelectrical impedance analysis
Densitometry	Body fat estimated from body density
Siri equation	Converts body density into body fat percentage
Fat-free mass	Same as lean body mass
Hydrophobic	Afraid of water

11. Where the sources differ

Topic	Lecture	Slides / E-book	What to do
BMI example	90 kg, 5 ft 8 in, height rounded to 1.7 m, BMI **31.03** (the spoken 1.7526 is wrong; 68 in is 1.7272 m)	Slides: 1.72 m, 2.95, **30.50**	Exact value is about 30.17. All are obese. Use the method, not the rounding
BMI categories	Asian criteria: obese at 25 and above	Slides: same. E-book: overweight 25 to 29.9, obese 30 and above	Use the Asian criteria taught in the course
Waist limits	90 cm men, 80 cm women	E-book obesity risk table: 102 cm men, 88 cm women	Use 90 and 80 for Indians and Asians
WC site	Across the iliac crest and belly button	Slides: umbilicus level and directly above the iliac crest	Same region: widest part of the waist
Skinfold sites	Abdomen 1 inch from the belly button	E-book: abdomen 2 cm to the right of the umbilicus	Use the lecture and slides
Gold standard	Not named in the lecture	Slides and E-book: hydrostatic weighing	Confirm in Fitness Testing 2
Order of testing	Six steps incl. muscular endurance	E-book: five steps (no separate muscular endurance)	Use the six steps
BIA prerequisites	Empty stomach, no exercise 12 h, no alcohol 48 h	E-book: no food or drink for 4 h, also void the bladder, avoid diuretics	Follow the stricter rule if unsure
BP three-day rule	Re-check over up to three days	Not in slides. E-book: average of at least 3 readings	Keep the lecture's rule for high BP on the day
Bod Pod accuracy and cost	Comparable to gold standard, expensive, not available in India	E-book: plus or minus 3%, 5 to 8 min, 30,000 to 40,000 US dollars	Treat figures as approximate

12. Quick revision questions

1. Name the two global and the four client objectives of fitness testing.
2. What is a test norm?
3. Why are the words "areas to work on" used instead of "weaknesses"?
4. What must the PAR-Q show before testing, and what is needed if there is one Yes?
5. What is the difference between the PAR-Q and the informed consent form?
6. List the six tests in order.
7. Why is blood pressure tested first, and what is done if it is high?

8. Why does the muscular strength test come before the muscular endurance test?
9. Why is flexibility tested last?
10. Why is the cardiorespiratory endurance test done before strength, endurance and flexibility tests?
11. What are the normal systolic and diastolic ranges, and how is a reading of 115/75 classified?
12. Where is the blood pressure cuff placed, and how tight should it be?
13. What does the waist circumference test tell you, where is the tape placed, and what are the male and female limits?
14. A woman has a waist of 30 inches and a hip of 36 inches. Calculate her WHR and give the risk.
15. Calculate the BMI of a person weighing 63 kg and 1.75 m tall, and name the category (Asian criteria).
16. Give two shortcomings of BMI.
17. What does a skinfold caliper measure, and what does a standard error of plus or minus 3% mean for a true value of 20%?
18. Expand BIA, state its principle and give the water percentage of muscle and fat.
19. List the BIA prerequisites with their time gaps.
20. How does the Bod Pod find body volume, and why is a swim cap worn?

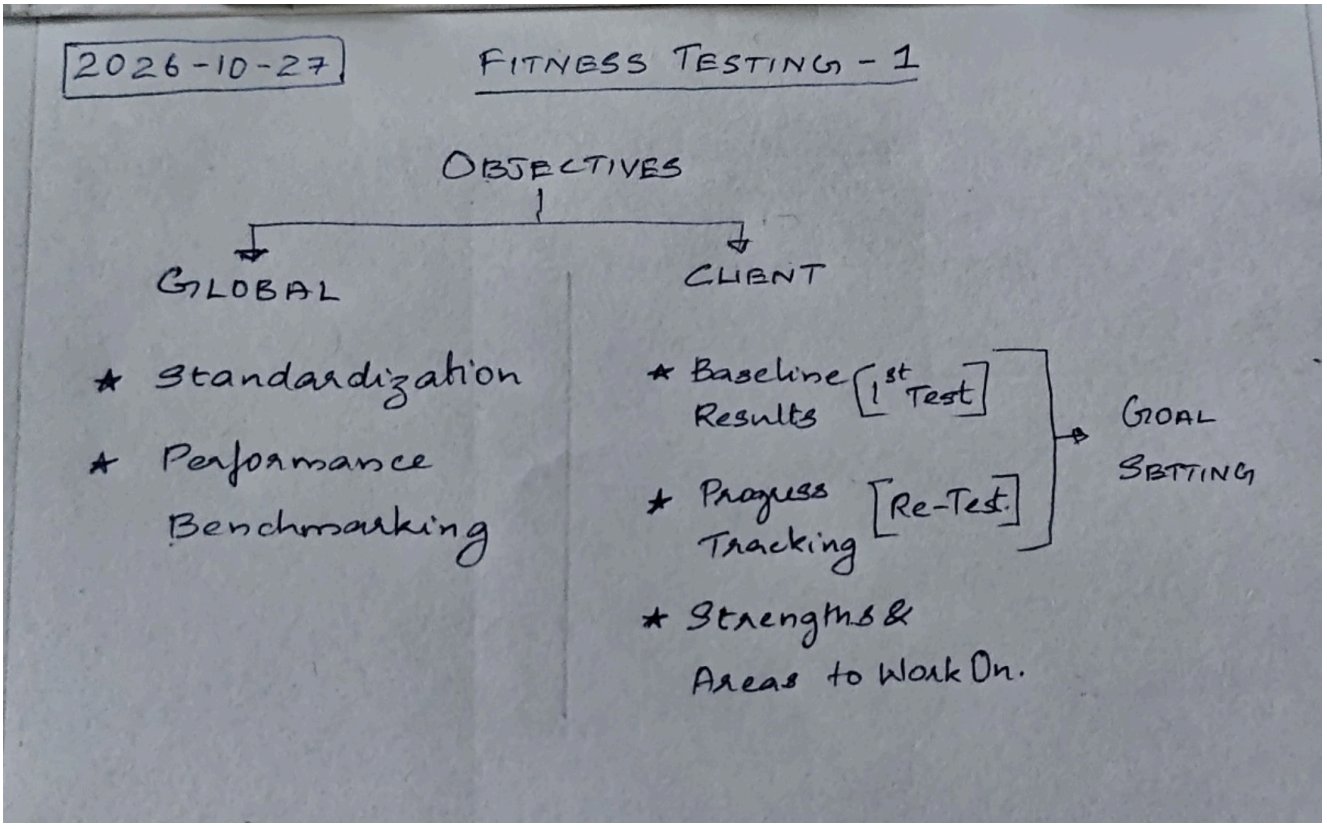
Answers

1. Global: standardisation and performance benchmarking. Client: baseline results, progress tracking, goal setting, strengths and areas to work on.
2. A reference sheet used to check a test score and see where the person stands for that fitness component.
3. A client meeting the trainer for the first time may react badly to being told they have "weaknesses", and the service industry needs polished wording. Strength here can come from any of the five components.
4. All answers must be No. With a Yes the trainer stops and asks, and the client must bring a NOC from their doctor.
5. The PAR-Q is a set of Yes or No medical questions that screens readiness. The informed consent form is a script that informs the client about machines, exercises and risks, ending with willing agreement and a signature.
6. Blood pressure, body composition, cardiorespiratory endurance, muscular strength, muscular endurance, flexibility.
7. It is a health marker: if BP is elevated or high, activity during testing could push it up dangerously. If high, the client rests, drinks water and is re-checked. If it stays high the test is skipped and tried again on the next days; high on three days means referral to a doctor.
8. A fatigued muscle cannot produce maximum force. Strength first also makes the endurance test feel easier (priming or relativity effect), so more repetitions may be done.
9. Warm tissues lengthen better. After the other tests the tissues are warm, so flexibility results are best.
10. Heart rate is the measure, and strength, endurance and stretching tests would also raise it, so the rise must come only from the cardio task.
11. Systolic 90 to 120 mmHg, diastolic 60 to 80 mmHg. 115/75 is within both, so it is normal BP.

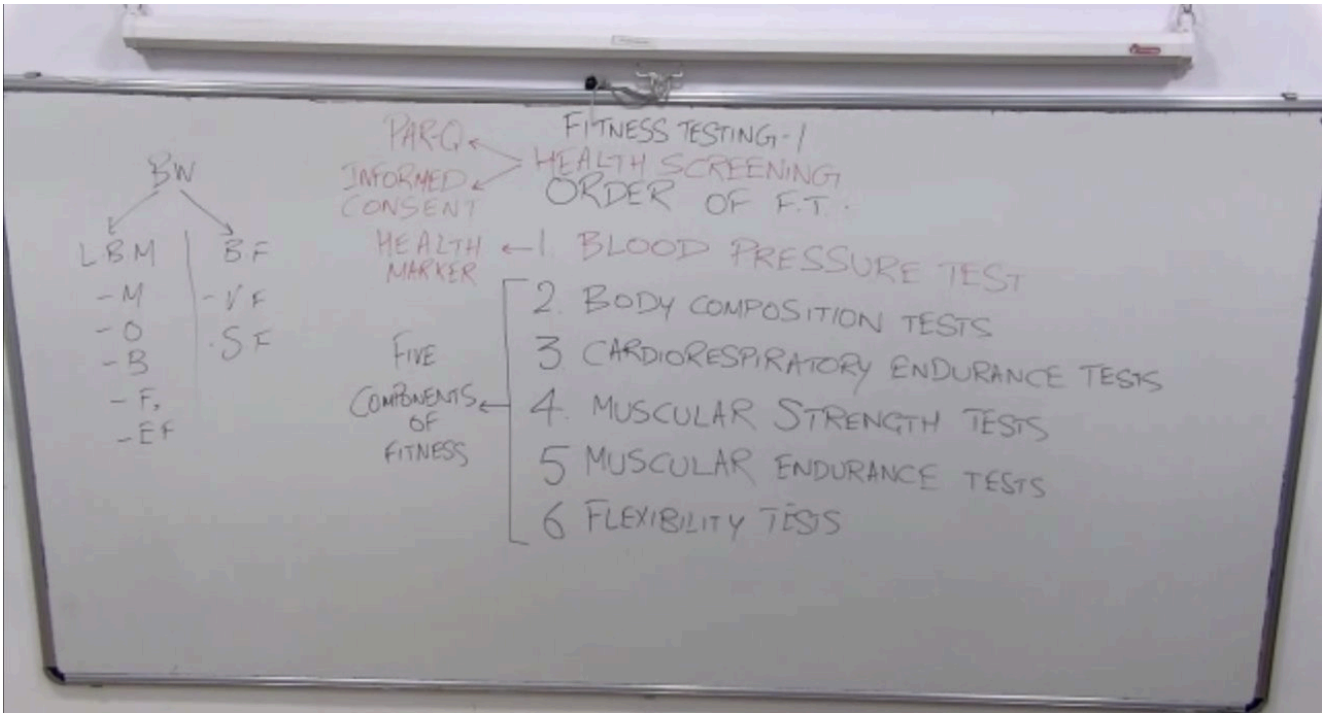
12. On the upper arm, with the edge 2 to 3 cm above the elbow, snug but not tight (two fingers fit under it), tube on the medial side, cuff at heart height.
13. It tells whether there is a health risk from visceral fat (central obesity). The tape goes across the iliac crest and belly button (widest part of the waist). Low risk: men below 35.5 in (90 cm), women below 31.5 in (80 cm).
14. WHR = 30 / 36 = 0.83. For a woman the limit is below 0.86, so it is low overall health risk.
15. 63 / (1.75 × 1.75) = 63 / 3.0625 = 20.6 kg/m², which is normal (18.5 to 22.9).
16. It does not separate body fat from lean body mass (a muscular person can be called obese), and it misses skinny fat people (normal BMI but fat in the belly).
17. It measures the thickness of a skin fold, which is subcutaneous fat (no visceral fat). A true 20% may read anywhere from 17% to 23%.
18. Bioelectrical impedance analysis. Water is a good conductor of electricity, so current passes smoothly through muscle (75% water) and is restricted by fat (10% water).
19. Rested and empty stomach (no water, food or coffee), no exercise for 12 hours, no alcohol for 48 hours, no metal objects, and women not during periods.
20. It subtracts the air volume of the chamber with the person from the air volume of the empty chamber (air displacement). A tight cap and swimwear stop air trapped in hair and clothes from entering the chamber.

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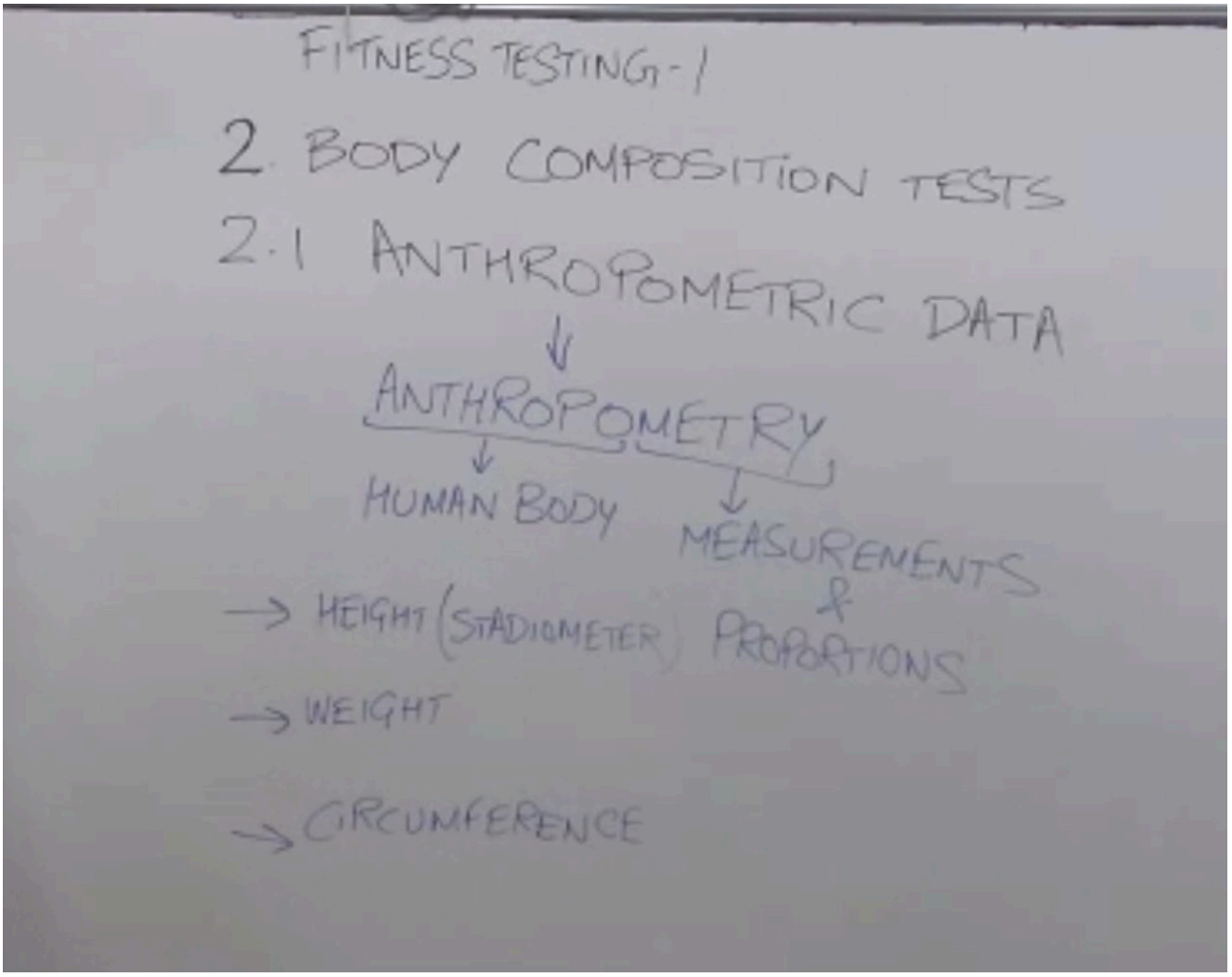
13. Board photos from the lecture



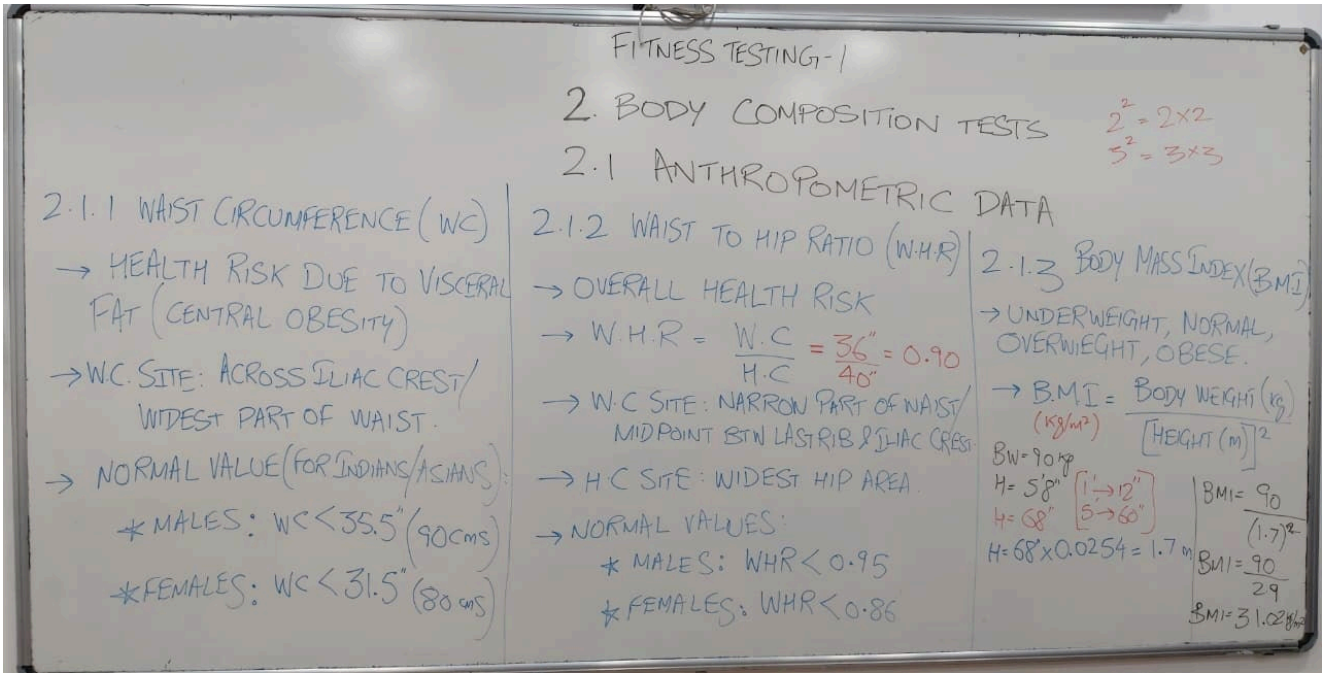
Objectives: global (standardisation, performance benchmarking) and client (baseline, progress tracking, goal setting, strengths and areas to work on).



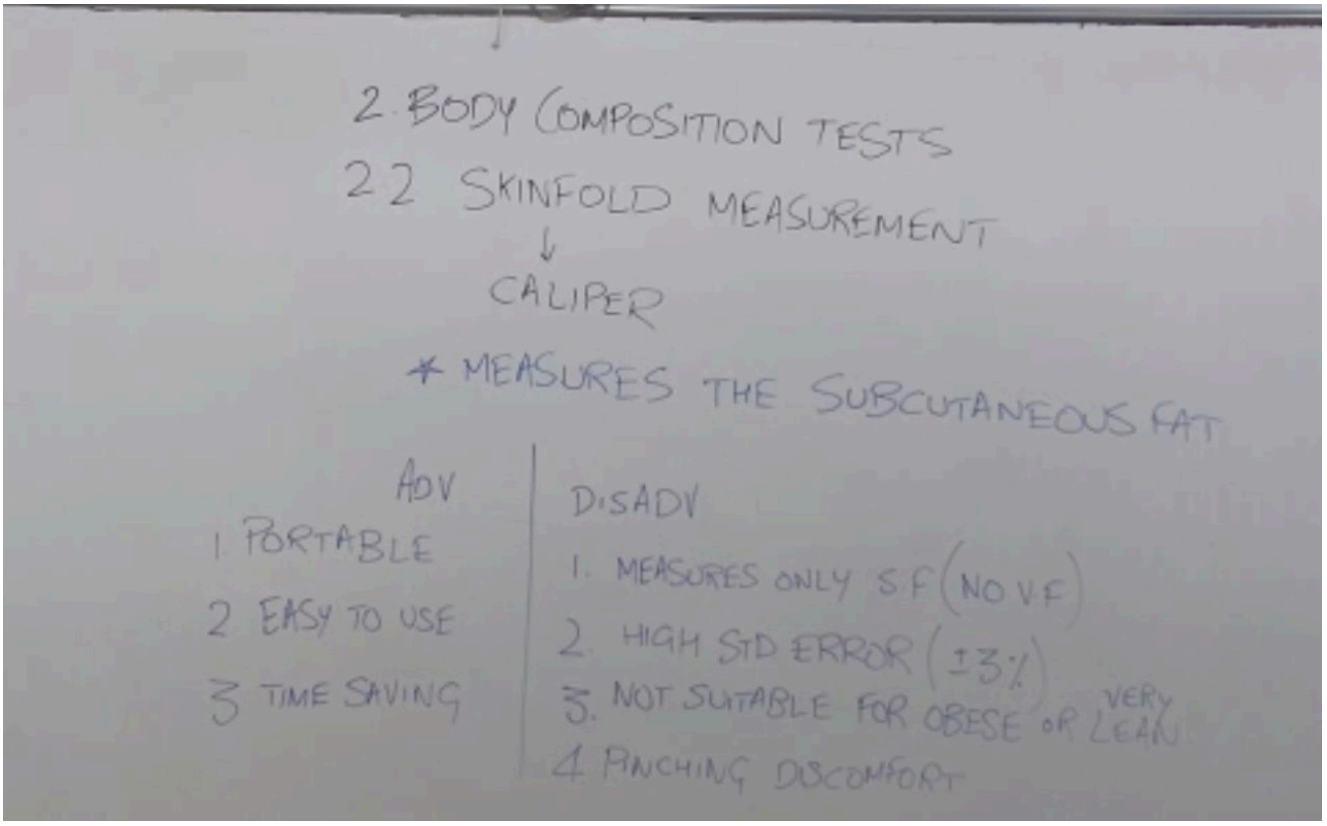
Health screening (PAR-Q, informed consent), the six-step order of testing, and the lean body mass and body fat memory aid.



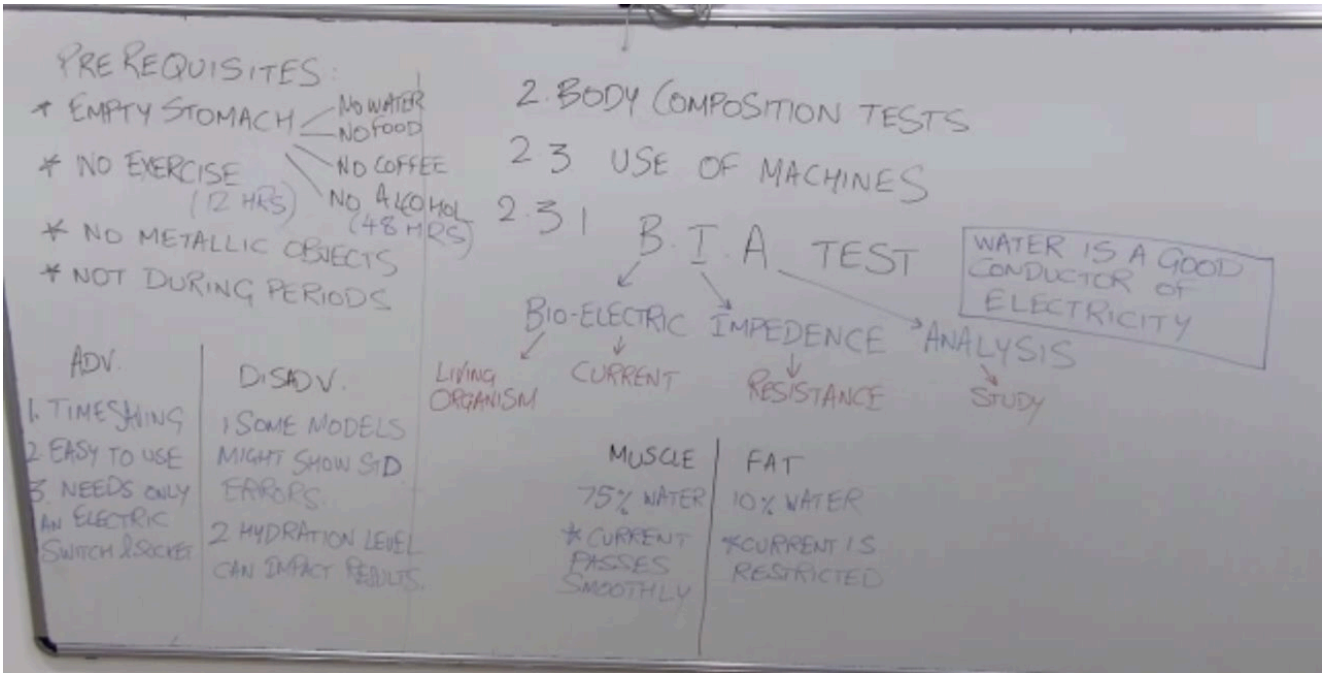
Anthropometry: human body, measurements and proportions; height (stadiometer), weight, circumference.



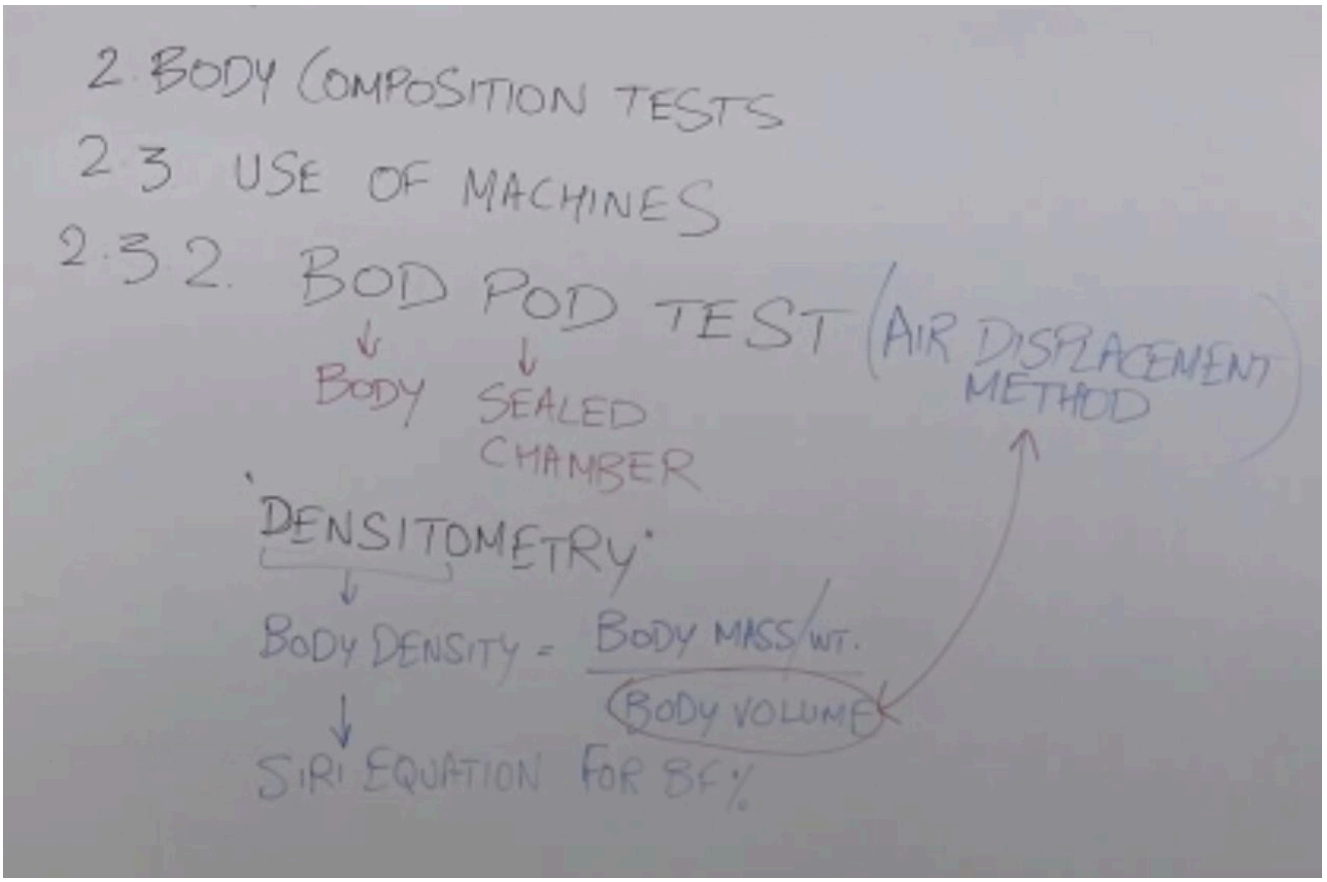
The three anthropometric tests with sites, norms and the worked WHR and BMI examples.



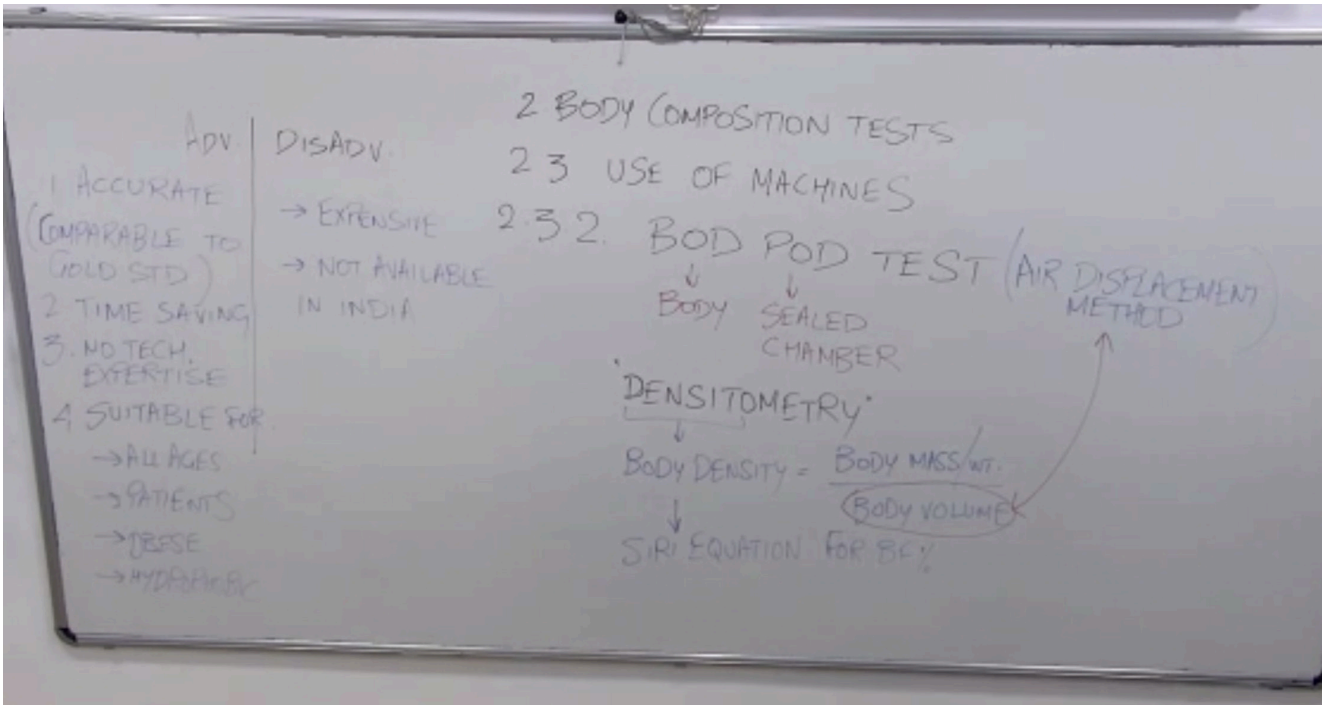
Skinfold caliper: measures subcutaneous fat, with advantages and disadvantages.



BIA: meaning of the name, muscle versus fat water content, prerequisites, advantages and disadvantages.



Bod Pod: densitometry, body density, volume by air displacement, Siri equation.



Bod Pod advantages and disadvantages (full board).